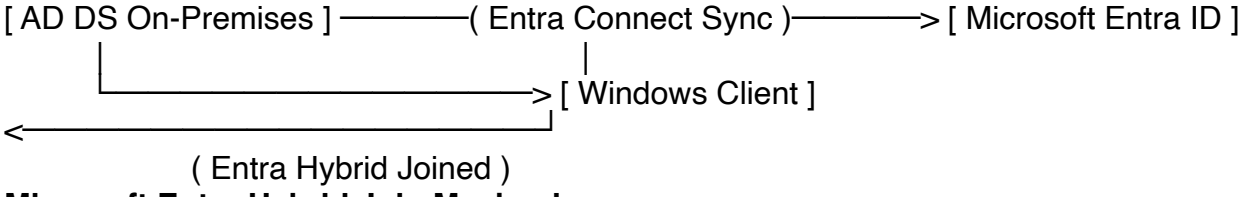
. Architecture & Low-Level Design (LLD)

Identity Foundation

Co-management requires a unified identity plane across on-premises Active Directory Domain Services (AD DS) and Microsoft Entra ID.

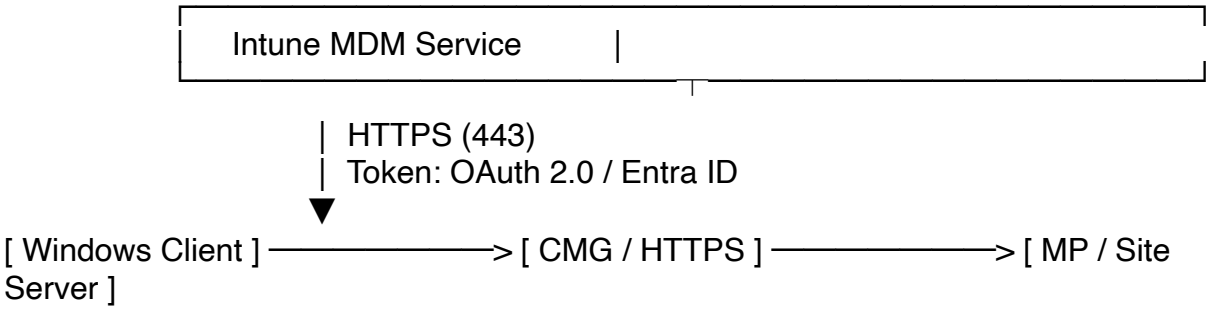


Microsoft Entra Hybrid Join Mechanics

- **Service Connection Point (SCP):** The client reads the SCP object from CN=62a0ed2e-66b3-4011-a684-362f20935351,CN=Services,CN=Configuration,DC=domain,DC=com via LDAP to retrieve azureADId (Tenant ID) and azureADName (Tenant Name).
- **PRT Generation:** Primary Refresh Token (PRT) issuance relies on valid Kerberos/ NTLM authentication to the On-Premises Domain Controller, combined with the device key pair generated during the Entra Hybrid Join flow (dsregcmd /status).
- **Synchronization Scope:**
    - **OU / Filter Scope:** Computer object Organizational Units (OUs) containing target ConfigMgr client devices must be explicitly included in the Entra Connect sync scope.
    - **Attribute Requirements:** Ensure userCertificate attribute is synchronized if using certificate-based device identification.
    - **User Principal Name (UPN):** User UPNs must resolve to a routable internet domain matching a verified custom domain in Entra ID (e.g., user@company.com, not user@company.local). If non-routable, configure an alternate UPN suffix in AD DS and update user accounts prior to MDM auto-enrollment.

Communication Channels

Clients maintain dual communication paths to ensure high availability and off-network management capabilities.



HTTPS (443) HTTPS (443)
Token: PKI / Mutual Auth PKI /
Entra ID / Token Server Cert

Protocols, Ports & Endpoints

- **ConfigMgr Client ↔ Management Point (MP):** HTTPS port 443 (or HTTP 80 if HTTP fallback is permitted; HTTPS strongly recommended). Mutual authentication via PKI client certificate or Entra ID token.
- **ConfigMgr Client —(Cloud Management Gateway / CMG)—> MP:** HTTPS port 443 to *.cloudapp.net or custom domain CNAME.
- **ConfigMgr Client ————————> Intune MDM Service:** HTTPS port 443 to *.manage.microsoft.com, *.omadm.manage.microsoft.com. Authenticated via Entra ID OAuth 2.0 user/device tokens.

Authentication Tokens & Certificates

- **Device PKI Certificate:** Used for mutual TLS to on-premises MPs and CMG when Entra ID token validation is unavailable.
- **Entra ID Device Token:** Acquired via Azure Device Registration Service (ADRS) during Hybrid Join; used for CMG authentication and Intune MDM enrollment.
- **Entra ID User Token:** Issued upon user login (backed by PRT); required to trigger user-scoped MDM auto-enrollment and evaluate conditional access policies.

Registry & CSP Mapping

Windows 10/11 utilizes Configuration Service Providers (CSPs) to enforce authority assignment locally.

Core Registry Keys

Co-management Configuration & Workload Bitmask:

Plaintext

HKLM\SOFTWARE\Microsoft\CCM\CoManagementFlags
Type: DWORD

Function: Stores the calculated workload bitmask received from the Management Point via client policy.

Enrollment Status:

Plaintext

HKLM\SOFTWARE\Microsoft\Enrollments\<EnrollmentID>
Key Values:

- EnrollmentState (DWORD): 1 = Enrolled, 0 = Not Enrolled.

- DiscoveryEndpoint (STRING): URL to the Intune discovery service.
- UPN (STRING): Enrolled user UPN.

DeviceManageabilityCSP:

Plaintext

HKLM\SOFTWARE\Microsoft\DeviceManageabilityCSP
Tracks active CSP policies applied by Intune versus policy overrides sent by ConfigMgr.

Authority Bitmask Values

ConfigMgr controls workload evaluation using an internal integer bitmask calculated by combining flags for individual management features.

Bitmask Table

Workload Name	Decimal Value	Hex Value	Managed Authority
Co-management Enabled	1	1	Core Co-management engine active
Compliance Policies	2	2	Intune evaluates device compliance
Device Configuration	4	4	Intune manages OMA-URI/CSPs
Resource Access Policies	8	8	Intune manages VPN/Wi-Fi/Cert profiles
Windows Update for Business	16	10	Intune manages WUfB rings
Endpoint Protection	32	20	Intune manages Defender/ BitLocker
Client Apps	64	40	Intune deploys Win32/M365 apps
Office Click-to-Run	128	80	Intune manages M365 Apps updates

Evaluation Rules

1. **Bitwise AND Operations:** CoManagementHandler.log reads CoManagementFlags. The client performs a bitwise AND check against incoming client settings to determine local agent behavior.
2. **Workload State Examples:**
    - 1 (0x00000001): Co-management enabled, all workloads remain with **ConfigMgr**.
    - 3 (0x00000003): Co-management enabled (1) + Compliance Policies shifted to **Intune** (2).
    - 255 (0x000000FF): All workloads fully shifted to **Intune**.

2. Step-by-Step Configuration Blueprint

Prerequisites & Permissions

- **MECM Infrastructure:** Version 2107 or higher (latest Current Branch recommended).
- **Service Connection Point:** Configured in **Online, persistent connection** mode.
- **Intune Licensing:** Microsoft 365 E3/E5, Enterprise Mobility + Security (EMS) E3/E5, or standalone Intune P1/P2 assigned to all target users.
- **Administrative Roles:**
    - **ConfigMgr:** *Full Administrator* scope on top-level site.
    - **Entra ID:** *Global Administrator* or *Privileged Role Administrator* (for initial app registration), plus *Intune Administrator*.

Tenant Onboarding

1. Open the ConfigMgr console, navigate to **Administration** > **Overview** > **Cloud Services** > **Cloud Attach / Co-management**.
2. Click **Configure Cloud Attach** to launch the wizard.
3. Select the cloud environment (**Azure Public Cloud**) and click **Sign In** with an Entra ID Global Admin account.
4. Select **Configure Co-management** options:
    - **Automatic enrollment in Intune:** Set to **Pilot** or **All**.
    - **Target Collection:** Specify the collection for auto-enrollment if **Pilot** is selected.
5. Grant admin consent for the required Entra ID Enterprise Applications created during setup:
    - **ConfigMgr Server App** (Web App / API)
    - **ConfigMgr Client App** (Native Client)

Enrolment Paths

Path 1: Auto-enrolling Existing Configuration Manager Clients

Configuration Steps:

1. In the Entra ID portal, navigate to **Microsoft Entra ID** > **Mobility (MDM and MAM)** > **Microsoft Intune**.
2. Set **MDM user scope** to **Some** (and select a target group) or **All**.

3. In the ConfigMgr Console, under Co-management properties, set **Enable Automatic MDM Enrollment** to **Pilot** or **All**.

Group Policy vs. ConfigMgr Auto-Enrollment Conflict
Do NOT enforce the GPO policy Enable automatic MDM enrollment using default Azure AD credentials on devices that are already managed by ConfigMgr auto-enrollment. Doing so can cause dual-enrollment race conditions, resulting in duplicate entries or MDM Enrollment Error 0x8018002b.

Path 2: Modern Provisioning / Bootstrap via Windows Autopilot

1. Provision device via Windows Autopilot (User-Driven or Self-Deploying).
2. During OOBE, Intune applies an Enrollment Status Page (ESP) policy configured to install the **ConfigMgr Client Bootstrap Package**.
3. Command Line syntax inside Intune Win32 App wrapper:

DOS

```
ccmsetup.exe /mp:https://CMG.CloudApp.Net/CCM_Service_IO/72057594037927939
CCMHOSTNAME=CMG.CloudApp.Net/CCM_Service_IO/72057594037927939
SMSSiteCode=PRI AADTENANTID=88888888-8888-8888-8888-888888888888
AADCLIENTAPPID=77777777-7777-7777-7777-777777777777
AADRESOURCEURI=https://ConfigMgrService
```

Pilot Collections & Phased Rollouts

Structure collections in ConfigMgr hierarchy using explicit query rules to isolate ring stages:

```
[ All Desktop Clients ]
        |
        ├──────> [ CoMGMT-Pilot-Stage1-IT ] (5% Devices)
        |
        ├──────> [ CoMGMT-Pilot-Stage2-EarlyAdopters ] (15% Devices)
        |
        └──────> [ CoMGMT-Prod-Broad ] (Remaining 80% Devices)
```

1. Navigate to **Asset and Compliance** > **Device Collections**.
2. Create pilot collections for each logical stage (e.g., CoMGMT-Pilot-Stage1-IT, CoMGMT-Pilot-Stage2-EarlyAdopters).
3. Under Co-management properties in Cloud Services, bind individual workload sliders to these specific pilot collections.

Workload Shifting Strategy

Workloads should be migrated in distinct phases to prevent policy conflicts.

Phase 1: Compliance Policies ——> Phase 2: Endpoint Protection ——> Phase 3: WUfB & Apps

WORKLOAD MIGRATION TIMELINE

Phase 1: Foundation
├── Compliance Policies (Intune evaluates access; ConfigMgr runs scripts)
└── Resource Access Policies (Wi-Fi, VPN, Certificate profiles)
    |
Phase 2: Security & Software
├── Endpoint Protection (Defender, BitLocker, Firewall)
└── Office Click-to-Run Apps
    |
Phase 3: Core Operations
├── Client Apps (Win32 & LOB deployments)
├── Device Configuration (CSPs replace GPOs)
└── Windows Update for Business (Quality & Feature rings)

Policy Conflict Mitigation Checklist

- **Windows Updates:** When moving WUfB to Intune, ensure ConfigMgr group policy **Disable dual scan** is NOT set to conflicting states. Intune configures the update policies directly via the Update Policy CSP.
- **Endpoint Protection:** Ensure local group policies or SCCM Antimalware policies do not re-apply settings over Defender policies deployed by Intune.

3. Comprehensive Troubleshooting Matrix & Log Analysis

Client-Side Logs

%WinDir%\CCM\logs\CoManagementHandler.log

Monitors workload state calculation and changes.

Plaintext

Processing workload updates...
Current Workload Flags: 1 (ConfigMgr native)
Merging Workload Flags from policy: 19 (0x00000013 -> Compliance, WUA shifted)
Successfully merged and calculated target state. Local bitmask updated to: 19.

%WinDir%\CCM\logs\ComplRelayAgent.log

Tracks translation of ConfigMgr compliance status up to the Intune Compliance CSP.

Plaintext

Relaying compliance state to Intune MDM channel...
Message payload formatted. Device ID: d67fa981-....
Successfully sent state status message to Cloud Relay. HTTP status 200 OK.

%WinDir%\CCM\logs\ClAgent.log

Processes Configuration Items; determines if compliance assignment belongs to SCCM or Intune.

Plaintext

CIAgentJob({AB123456-...}): Enforcing policy authority check.
DCM authority for CI [ScopeId_.../DeploymentType_...] is Intune. Skipping local enforcement.
%WinDir%\CCM\logs\WUAHandler.log

Monitors Windows Update Agent behavior when shifting the WUfB workload.

Plaintext

Group policy setting for Windows Update Server is disabled due to WUfB workload shift.
Enabling Dual Scan suppression / deferral control under Windows Update Agent.
Source manager indicates updates will be processed directly via Microsoft Update / Intune policy.
%WinDir%\CCM\logs\EndpointProtectionHandler.log

Tracks takeover of Endpoint Protection (Defender/BitLocker) by Intune.

Plaintext

Endpoint Protection workload flag is enabled in CoManagementFlags (Bit position 5 set).
Bypassing local ConfigMgr Antimalware Policy application. Intune CSP is active authority.

Registry & Event Viewer Checks

Key Registry Paths to Inspect

MDM Enrollment Status:

Plaintext

HKLM\SOFTWARE\Microsoft\Enrollments\FFFFFFFF-FFFF-FFFF-FFFF-FFFFFFFFFFFF
Check EnrollmentState = 1.

Active CSP Policy Evaluation:

Plaintext

HKLM\SOFTWARE\Microsoft\PolicyManager\current\device\ControlPolicyConflict
Look for conflicting policy flags registered between SCCM and Intune.

Entra ID Join Info:

Plaintext

HKLM\SYSTEM\CurrentControlSet\Control\CloudDomainJoin\JoinInfo
Event Logs

- **Path:** Applications and Services Logs > Microsoft > Windows > DeviceManagement-Enterprise-Diagnostics-Provider > Admin
- **Event ID 71:** MDM Enrollment succeeded.
- **Event ID 76:** MDM Enrollment failed (Inspect error code in event body).
- **Event ID 404:** MDM Enrollment failed due to invalid access rights / server side reject.

Common Failure Scenarios & Resolutions

Scenario A: Devices Stuck in Pre-Co-Managed State / Enrollment Fails

Diagnostic Indicators

- **Log:** CoManagementHandler.log shows Device is not Entra joined, skipping MDM auto-enrollment.
- **Event Viewer:** Event ID 76 with error code 0x8018002b ("User canceled operation") or 0x8007064a.

Root Cause

Device is not fully Entra Hybrid Joined yet, or user lacks an Intune license, or dual GPO auto-enrollment policy is interfering.

Resolution Path

1.Verify Entra Join Status: Terminal check.
Run dsregcmd /status from an elevated prompt. Confirm both AzureAdJoined : YES and PRT : YES.

Verification: Command output displays AzureAdJoined : YES and EnterpriseJoined : NO.

2.Verify Entra User Scope:
Open the Entra ID Portal, navigate to **Mobility (MDM and MAM)** > **Microsoft Intune**, and verify the target user is included in the MDM User Scope.

Verification: The user account is a member of the assigned Entra ID group.

3.Clear Stuck Enrollment Registry Keys: Registry cleanup.
If error 0x8018002b persists, delete the corresponding registration GUID key under:

HKLM\SOFTWARE\Microsoft\Enrollments\

Verification: The stale GUID key is removed, triggering a clean re-enrollment attempt.

4.Trigger Re-enrollment Task:
Run the scheduled task under Microsoft > Windows > EnterpriseMgmt > [EnrollmentID] > Schedule created by enrollment main for prompt customer.

Verification: Event ID 71 appears in the DeviceManagement-Enterprise-Diagnostics-Provider/Admin log.

Scenario B: Policy Conflicts & Dual Scan Issue for Windows Updates

Diagnostic Indicators

- **Log:** WUAHandler.log reports Group Policy set to use WSUS, but WUfB policy applied. Dual scan active.
- Devices scan against WSUS and Microsoft Update simultaneously, causing unexpected updates or broken patching schedules.

Root Cause

The WUfB workload slider was moved to Intune, but local Active Directory GPOs still enforce explicit WSUS endpoints (Specify intranet Microsoft update service location) without dual-scan controls configured correctly.

Resolution Path

1.Check WSUS Policy Enforcement:
Run gpresult /h gpreport.html to identify GPOs deploying Specify intranet Microsoft update service location.

Verification: Report identifies the specific GPO linked to the OU.

2.Configure Dual Scan Suppression:
Set the GPO setting **Do not allow update deferral policies to cause scans against Windows Update** to **Disabled** or set **Disable Dual Scan** via ConfigMgr Client Settings to **Yes**.

Verification: WUAHandler.log confirms WSUS endpoint policy is managed via local CSP policy fallback.

Scenario C: Workload Toggle Issues (Client Refuses Switch)

Diagnostic Indicators

- Console shows slider moved to Intune, but CoManagementHandler.log reports WorkloadFlags match local policy, no change.
- HKLM\SOFTWARE\Microsoft\CCM\CoManagementFlags does not update to match the console bitmask.

Root Cause

The client has not received the updated policy from the Management Point, or the client is targeted by conflicting Co-management settings from multiple collections.

Resolution Path

1.Verify Policy Download: Client action.

Force a policy pull on the client via **Configuration Manager Control Panel** > **Actions** > **Machine Policy Retrieval & Evaluation Cycle**.

Verification: PolicyAgent.log records successful receipt of new policy package.

2.Check Resultant Workload Target:
Open PowerShell and run:

PowerShell

Get-CimInstance -Namespace "root\ccm\dcm" -ClassName "CCM_CoManagementState"
Verification: Returned WorkloadFlags match the calculated bitmask configured in the SCCM console.

Scenario D: "Tattooing" Effects When Reverting Workloads to ConfigMgr

Diagnostic Indicators

- Slider moved back from Intune to ConfigMgr, but Intune CSP settings (e.g., Password Complexity, BitLocker encryption keys) remain active on the device.

Root Cause

Intune MDM CSPs are persistent by default ("tattooing"). When authority shifts back to ConfigMgr, Intune does not automatically purge applied CSP registry entries unless explicitly unassigned prior to shifting.

Resolution Path

1.Unassign Intune Profiles:
Before pulling the workload slider back to ConfigMgr, exclude or unassign the Intune configuration profiles from the target device collection in the Intune portal.

Verification: Intune reports configuration status as **Not Applicable** for the target devices.

2.Manually Clear Persistent CSP Entries: Remediation script.
Run a remediation script to clean up orphaned entries under:

HKLM\SOFTWARE\Microsoft\PolicyManager\current\device\

Verification: Registry path no longer contains Intune-specific CSP values.

3.Re-enforce ConfigMgr Compliance Baseline:
Deploy a Configuration Baseline in ConfigMgr to re-assert the required setting state.

Verification: CIAgent.log confirms local DCM baseline enforcement is active.